

GE261004 DESIGN THINKING

Unit V – Innovation, Entrepreneurship and Sustainability · Presentation 6 of 6

Ethical Considerations in Design

Every design is a decision about somebody · the engineer's code · bias and privacy · dark patterns · three questions before you ship

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In this presentation

- Every design decides about somebody
- The engineer's code
- Four ways design causes harm
- Bias in design
- Privacy and data
- Dark patterns
- Persuasion and manipulation
- When you are told to do it anyway

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The designer's responsibility

You are accountable for what your design does, not only for what you intended

Every design is a decision about somebody

- Who it serves, who it excludes and who pays are design decisions, not side effects
- A design that works only for people like you has made a decision about everyone else
- “We did not think about that” describes how the harm happened; it does not excuse it
- The people affected are rarely the people in the room when it is decided

KEY POINT If you have not decided whom your design serves, you have still decided. You just did not watch yourself do it.

The engineer's code

- The Institution of Engineers (India) asks members to use their knowledge for the welfare, health and safety of the community, without discrimination
- To work only within their competence, with care, sincerity and honesty
- To be honest and fair in public statements, on the basis of adequate knowledge
- To inform employers and clients of the environmental, economic and social consequences of their work

KEY POINT Telling the client what the consequences are is in the code. Staying quiet because nobody asked is not a defence.

Four ways design causes harm

Exclusion

It works, but not for a whole group of people – by cost, ability, language or access.

Deception

It works by misleading: hidden costs, hidden defaults, claims nobody measured.

Extraction

It takes more than the user knowingly gave – data, attention, money, time.

Unsafe use

It behaves as designed but fails the person using it in the real, untidy world.

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Bias, privacy and persuasion

Three places harm arrives without anybody intending it

Bias in design

- Any design carries the assumptions of whoever made it, including the unexamined ones
- Data-driven systems inherit whatever bias is in the data they were trained on
- Test with people unlike you – it is the only reliable way to find your assumptions
- Ask who is missing from the test group, every time, and write the answer down

KEY POINT You cannot inspect your way out of your own assumptions. You can only test your way out.

Privacy and data

- Collect only what the design actually needs, and say plainly what it is for
- Consent means the person understood – not that they clicked past a wall of text
- Data you never collected cannot be stolen, leaked or subpoenaed
- The people in your field notes trusted you; anonymise them and keep it that way

KEY POINT The default should be not collecting it. Make somebody argue for each field, rather than argue against it.

Dark patterns

Roach motel

easy to join,
very hard to leave

Confirmshaming

“no thanks, I like
wasting money”

Drip pricing

fees appear only
at the last step

Misdirection

the bright button is
the one you did not want

Forced continuity

the free trial bills
you without warning

Privacy by default

sharing is on unless
you go and find it

Harry Brignull named these in 2010; several are now unlawful in several jurisdictions.

Persuasion and manipulation

- Persuasion gives the person a reason; manipulation exploits a weakness
- A default is a design decision – it is where most people will end up
- “It increased conversions” measures the effect, not whether it was right
- If the design only works while the user misunderstands it, it is manipulation

KEY POINT The test is simple: would it still work if the user understood exactly what was happening?

Three questions before you ship

Three questions before you ship

1

Would you tell the user
exactly what it does?

2

Would you accept it
being done to you?

3

Would you defend it
in public, by name?

any “no” is the design telling you something

Ask them out loud, in the room, before the decision is made.

When you are told to do it anyway

- Raise it early, in writing, in plain language, describing the consequence
- Offer an alternative – objections with options are far harder to dismiss
- Escalate through the proper channel, and keep your own record
- Know where your line is before you are standing on it

KEY POINT Deciding in advance what you will not do is the only preparation that works. Nobody has ever worked it out under pressure.

In class: the design dilemma debate

- Each team is given one real dilemma and a side to argue, whether they agree or not
- Four minutes to argue, two to respond, two for the room to question
- Then each team states what it actually believes, and why
- Finish by writing one line you would not cross in your own project

ACTIVITY Whole hour, structured debate. Arguing a side you disagree with is the point of the exercise.

THINK ABOUT IT

Where is the line you will not cross as an engineer – and have you ever written it down?

- What would you refuse to design?
- What would you refuse to stay quiet about?
- Who would you tell, and in what order?
- What would it cost you, and is that a price you accept?

Summary

- Who it serves, excludes and charges are design decisions, not side effects
- The IEI code asks for public welfare, competence, honesty and disclosure
- Harm arrives as exclusion, deception, extraction or unsafe use
- Bias is escaped by testing with people unlike you, not by inspection
- Collect only what the design needs; consent means understood
- If it only works while the user misunderstands it, it is manipulation

That is Unit V – and the course

- Revision of all five units, in the next session
- Final team pitches: three minutes, with your prototype
- One POV, one tested prototype, one evidence-backed change
- Thank you – and go and find a real problem